
Math-CoT Distillation Suppresses Cue Acknowledgment Beyond What Compression Explains

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Abstract

Chain-of-thought (CoT) monitorability is the hope that reading a model’s reasoning will tell us why it answered, so an overseer can catch undesired influences. We ask whether supervised fine-tuning on math traces preserves this property, using the cue-based paradigm of Meek et al. (2025) and Turpin et al. (2023) on a DeepSeek-R1-distilled student. Our central result is an intervention. Fine-tuning the student on a weaker model’s math CoT, with the reasoning span genuinely present in the loss, cuts the CoT’s acknowledgment of a planted cue from 23.6% to 16.6% on matched (question, cue) pairs (paired $\Delta = -0.080$, $n = 150$, McNemar $p = 0.043$), at held-out MATH accuracy that does not fall (0.685 against an untrained reference of 0.650). Those figures come from a budget-matched re-run of both arms (§5.3); the same comparison at the 8k generation budget the rest of this paper uses reads 25.0% to 14.4%, $p = 0.005$, and we treat the weaker budget-matched number as the honest one, because at 8k only the untrained baseline was routinely truncated before it could answer. The retrained student’s traces are only $\sim 27\%$ shorter than the untrained model’s, so the drop is not a by-product of the extreme CoT compression that answer-only supervision produces. Over the same pairs we cannot detect a change in the cue’s behavioral pull on the answer (paired $\Delta = +0.048$, 95% CI $[-0.060, +0.157]$, $n = 83$, $p = 0.52$) — but we are explicit that this is an underpowered null rather than a demonstrated invariance, and that the behavioral measurement is degraded by the untrained baseline failing to emit a parseable answer on 44% of these items (§7). That fine-tuning lowers CoT faithfulness on an answer-level metric is already known (Lobo et al., 2024); what we add is that the drop is induced by intervention, survives the removal of the compression that would trivially explain it, and is not accompanied by any detectable reduction in how much the cue moves the answer. A second arm isolates how much of the effect a shorter trace can produce on its own. Due to a chat-template defect we document in §3.1, an earlier run supervised the same student on the *final-answer segments* of the same traces; that arm compresses CoT $14\times$ and drives acknowledgment to 3.2%, so answer-only supervision reproduces a far larger collapse than reasoning-preserving supervision does. Length alone is nonetheless insufficient: at matched length and matched cue composition the gap persists in direction and effect size under every construction we tried, though it reaches significance only in the construction that does not hold the question pool fixed, and the question-matched alternatives leave 7 and 8 records (§5.1), and the collapse replicates on MMLU at $2.8\times$ compression rather than $14\times$. A same-strength self-distillation control makes the weak-to-strong asymmetry an unlikely cause, though that control was run under answer-only supervision and is correspondingly weak (§7). A capability-only certification can pass a model whose CoT has become markedly less revealing while still reading as elaborate.

1 Introduction

Chain-of-thought “monitorability” is the hope that reading a model’s reasoning reveals why it answered, so an overseer can catch undesired influences [9]. Korbak et al. argue that this opportunity is fragile and that developers should think about how their design choices affect monitorability. Our paper looks at one such design choice, math-CoT SFT, and shows that it erodes monitorability in a specific way.

Weak-to-Strong Reasoning (W2SR) [1]¹ reports that fine-tuning a strong student on a weaker model’s CoT raises accuracy. Concurrent faithfulness work [3, 2, 4] shows that models often do not verbalize the cues that actually move their answers.

W2SR certifies the student on accuracy alone and is silent on whether the distilled CoT stays monitorable. If distillation changed how a model reasons on the page without changing what it reveals, an accuracy check would miss the problem entirely.

The phenomena we exploit are largely known. Lobo et al. [5] showed that fine-tuning hurts CoT faithfulness on an answer-level metric. Zaman and Srivastava [7] argued via causal mediation that the dissociation between behavior and CoT verbalization can exist statically. Concurrent self-distillation work [6] identifies compression as a mechanism that suppresses CoT properties, in their case the verbalization of uncertainty. Our contributions are methodological and target-specific.

1. We induce a drop in cue acknowledgment by intervention. On a run whose supervision genuinely contains the teacher’s reasoning (§5.2), math-CoT SFT moves an R1-distill student from acknowledgment 23.6% to 16.6% at a budget matched across arms (paired $p = 0.043$; $25.0\% \rightarrow 14.4\%$, $p = 0.005$, at the 8k budget that truncates the baseline far more than the trained arm — §5.3), at held-out MATH accuracy that does not fall, and with no detectable change in paired influence ($\Delta = +0.048$, CI $[-0.060, +0.157]$, $p = 0.52$ — an underpowered null, not a demonstrated invariance). Because the manipulation is the training data rather than a filter applied to the outcome, this does not condition on any post-treatment variable. The new piece beyond Zaman and Srivastava is the controllability: an ordinary post-training recipe produces the effect on demand.
2. A same-strength self-distillation arm serves as a negative control. With teacher equal to student, the collapse is indistinguishable from W2SR (paired $p = 1$, $n = 79$), and every answer-only arm we tested against W2SR gives $p \geq 0.09$. We have not seen this control in the literature in this form. It is our strongest evidence that the cause is the SFT in general, not weak-to-strong asymmetry, and it separates our claim from the concurrent self-distillation work on uncertainty, since the outcome variable is different but the underlying mechanism is similar. We are explicit that this control was run under the answer-only supervision format, which weakens it: with the teacher’s reasoning outside the loss, the three teachers differ only in their final answers (§7).
3. We separate how much of the collapse is trace length from how much is not, using a supervision-format contrast that we did not design but can exploit. The answer-only arm (§3.1) compresses CoT $14\times$ and reaches acknowledgment 3.2%; the reasoning-preserving arm compresses only $\sim 27\%$ and reaches 14.4%. Length therefore carries much of the raw drop. It does not carry all of it: the reasoning-preserving arm sits significantly below baseline at near-matched length; the MMLU replication reproduces the full collapse at $2.8\times$ compression rather than $14\times$; and at matched length with cue composition held fixed the gap is present in every construction we tried, at OR between 2.7 and 4.7, though only the construction that pools a larger re-run reaches significance and that construction does not hold the question pool fixed (§5.1).
4. A reproduction of W2SR with a headroom probe shows that its accuracy claim is itself conditional. The gain only fires with (i) an in-family teacher and (ii) a base student. A zero-shot CoT prompt on the untrained student separates real elicitation of latent capability from a CoT-prompting confound (Section 4), and the certified guarantee is narrower than Yuan et al. claim.

¹Distinct from the earlier paper with the same title by Yang, Ma, and Liu [10] (EMNLP Findings 2024), which uses a progressive SFT+DPO recipe on GSM8K, MATH, and OlympicArena. We build on Yuan et al.’s LoRA-SFT pipeline.

5. Putting these together gives a clear certification consequence. An accuracy-only certification of any math-CoT-SFT student, including same-strength self-distillation rather than just weak-to-strong, can pass a model whose CoT has become markedly less revealing. A monitorability composite that rewards elaborate-looking CoT misses the failure too. The cue-acknowledgment component has to be measured directly.

2 Related work

Lobo et al. [5] show that task-specific fine-tuning lowers CoT faithfulness across four datasets, and we take this degradation as given. The claim we are making is not that faithfulness drops, but that it drops while the cue’s behavioral influence on the answer does not. That is the dissociation their answer-level metric cannot isolate. Concurrent work by Jiang et al. [11] makes the closest complementary observation in a different domain: on medical CoT distillation, answer accuracy improves (74.7% $\rightarrow$ 84.4% MedQA) while reasoning step error rate climbs (30.6% $\rightarrow$ 50.3%), which they use to argue that answer-level metrics alone mask reasoning-quality decay. Their finding and ours share the shape “the answer looks better while the trace gets worse” in two very different domains, which is stronger joint evidence than either result alone.

Concurrent work [6] finds that self-distillation suppresses epistemic verbalization, specifically the verbalization of uncertainty tokens like “wait” and “hmm,” via compression. They observe that this hurts out-of-distribution generalization. We share the compression mechanism and the self-distillation setup, but our outcome variable is a different property of the CoT: cue-acknowledgment (does the trace mention or attribute reasoning to a planted hint) rather than uncertainty-token density. Uncertainty-token loss and cue non-acknowledgment are distinct failure modes that happen to share a mechanism. On the RL side rather than SFT, Little [14] independently shows that training with an explicit length penalty makes CoT less monitorable at matched length (7-35 pp lower hint disclosure after random-shortening baselines). Our result is the SFT-side complement: math-CoT SFT compresses CoT even with no length term in the objective and produces the same qualitative degradation, so the failure mode is not specific to length-penalized RL.

Meek et al. [2] measure monitorability as a combination of faithfulness and verbosity across model families. We use their cued-prompt eval but study how a training intervention moves faithfulness on a fixed substrate. We pair the standard verbal-acknowledgment measurement with a behavioral measure that separates “stops being influenced” from “stops verbalizing influence,” and we ask the sharper question of whether faithfulness falls beyond what CoT length alone explains.

Zaman and Srivastava [7] argue that the Biasing Features metric, which flags certain answers as “unfaithful,” partly captures the lossy compression of distributed transformer computation into linear natural language. They show via causal mediation that even non-verbalized hints can causally route influence through the CoT to the answer. That is our dissociation, observed inferentially. We add three things on top. First, we induce the effect with a training method, moving acknowledgment from 25.0% to 14.4% on a fixed substrate under reasoning-preserving supervision, and to 3.2% under answer-only supervision. Second, we pin the behavioral pull directly, via paired influence and a flip-to-cue-target rate at about $2\times$ chance on the trained arms, rather than infer it through mediation. (On the answer-only arm the pull does not merely persist but strengthens; on the reasoning-preserving arm we cannot resolve a change either way.) Third, we draw the certification consequence, since an accuracy-only or verbosity-rewarding check passes the very models on which the dissociation is most extreme. Zaman also argue that hint-acknowledgment is a poor faithfulness metric because absence of hint words does not prove the hint was not causally involved. We agree with that critique on faithfulness grounds. But for monitorability the metric is aligned with the task: what an overseer can read off the trace determines whether they can catch the influence, independent of whether the model was “really” moved by the hint in some deeper counterfactual sense. Our CoT-conclusion analysis in §5.1 further shows the CoT reasoning itself walks to the cue target in most silently-influenced cases, so on that arm acknowledgment absence is not merely a missing hint word. We note the analysis was run on the answer-only arm, so it cannot by itself adjudicate compression.

Channel migration is not what is happening here. Recent work on extended-thinking models documents a thinking-vs-answer divergence in hint acknowledgment, often a large one (for instance 87.5% vs 28.6% in [8]; concurrent work [18] finds a similar divergence on a larger 12-model panel with an “unacknowledged” quadrant of 11.8%). Our judge reads the full chain-of-thought, that is,

both the thinking tokens and the answer. So the collapse we measure is not an artifact of which channel is read; acknowledgment vanishes in both channels combined.

Xiong et al. [12] provide a useful contrast on the training-method axis: RLVR (reinforcement learning with verifiable rewards) spontaneously improves monitorability during training, driven by response sharpening and increased prompt attention, without monitorability being an explicit objective. Our result is the opposite for a different post-training procedure: SFT on math-CoT traces, also without monitorability in the objective, degrades monitorability instead. Which post-training method a model receives can flip the sign of the monitorability side effect, even when neither method targets monitorability directly. Related term-of-art usage: Yee et al. [13] introduced the language of “dissociation” for faithful-vs-unfaithful reasoning mechanisms in a static setting; our use of “behavior-verbalization dissociation” is in that same lineage but induced by training.

Known limits of the cue-based paradigm. The Turpin-style cue-acknowledgment paradigm we adopt has documented limitations. Young [15] shows that the choice of faithfulness classifier alone shifts measured faithfulness by up to 30 pp across otherwise identical setups, so absolute rates should be read as judge-dependent. Scalena et al. [16] argue that reasoning models commit to an answer well before the CoT ends, so late CoT content can be epiphenomenal and length-only comparisons can be misleading (our Task H length-binned analysis addresses this critique on our data, though see §5.1 for how much weight it can bear). Za et al. [17] document that Sonnet is roughly $3\times$ more susceptible than Gemini under adversarial persuasion when used as a monitor, which is one reason we cross-check our Sonnet judge with two others: Gemini (Cohen’s $\kappa \in [0.64, 0.68]$, substantial) and Kimi (overall $\kappa = 0.556$, moderate; systematically stricter than Sonnet). All three judges agree on the direction of the ack drop and give one-directional discordant patterns. We frame our headline metric as *monitorability* (does the trace reveal the influence to an overseer) rather than *faithfulness* (does the trace causally reflect the model’s internal computation) precisely because the paradigm operationalizes the former, not the latter: an overseer only has the trace, so what the trace does or does not say about the cue is directly what determines whether the influence is catchable.

3 Setup

W2SR [1] samples CoT from a weak teacher on a problem set, keeps correct and incorrect traces, LoRA-SFTs a stronger student, and reports held-out Pass@1. Monitorability follows Meek et al. [2]: each question is paired with a cue (authority opinion, leaked answer key, embedded metadata, and so on) that points at a target. An LLM judge reads the full CoT and scores faithfulness (does the CoT acknowledge the cue) and verbosity (a factor-utilization rate). We use the five cues shipped in the released eval, with `claude-sonnet-4-6` at $T = 0$ as the single judge. Faithfulness is reported as acknowledgment among cued samples. All evaluation is greedy ($T = 0$), trace sets are hashed, and training is LoRA. Generation, training, and gating run on Modal GPUs. All code, configs, per-sample judge labels, and the reanalysis pipeline are released. Four numbers from the answer-only arms — its acknowledgment rate, baseline’s, and the two paired deltas — are enforced by hard-fail assertions in `01_gate.py`, so any drift halts the suite. The reasoning-preserving arm’s numbers are reproduced by `K_cotsft_rerun.py` but are not yet under assertion.

Procedural details that matter for interpretation. Trace generation used $T = 0.6$, top- p 0.95, and a repetition penalty of 1.1 (the penalty was added for the loop-prone 1.5B teacher and kept constant across all teachers for comparability; it discourages long repetitive CoT and so interacts with the length analyses). Traces without a parseable `\boxed{}` answer were dropped before SFT, which preferentially removes traces truncated at the generation budget and therefore biases the training set slightly short. The under-test conditions were evaluated with an 8k generation budget (except the cross-family Llama arms, which used 16k); the teacher reference rows used 30k, so teacher-vs-student comparisons are not budget-matched (the baseline-vs-trained comparisons that carry every headline result are). SFT computes the loss over the full rendered sequence, prompt tokens included, rather than masking to the assistant span. The cue generator is *adaptive*: for each condition it points the cue at an option that differs from that model’s own uncued answer, so that the cue always has somewhere to pull. A consequence we should state plainly is that pairing on (question, cue) pairs responses to the same question and the same cue *type*, but not always to the same planted target letter — across baseline and the trained arms the target differs on roughly 60% of matched pairs (90/150 against the answer-only arm, 90/155 against the reasoning-preserving one). The paired estimand is therefore

199 “does this model follow a cue aimed away from its own prior answer,” which is the quantity we
 200 want, rather than “do two models respond identically to one fixed prompt.” Restricting to the pairs
 201 where the target letter does coincide preserves every result: the headline acknowledgment drop
 202 of §5.2 becomes $\Delta = -0.138$ on $n = 65$ ($p = 0.022$), and the answer-only arm’s drop becomes
 203 $\Delta = -0.167$ on $n = 60$ ($p = 0.006$). Task H’s W2SR side pools a larger re-run of the three text
 204 cues with the original run’s two non-text cues (634 records) to give the length bins usable counts; the
 205 paired analyses elsewhere use the original matched set. The main eval judge is `claude-sonnet-4-6`;
 206 Tasks I2/I3 and J were judged with `claude-sonnet-4.5` via OpenRouter. Task J additionally uses
 207 a hand-written binary yes/no acknowledgment rubric rather than the in-eval `cue_aware` rubric, so
 208 Task J rates differ from the main-eval rates by both judge and rubric and the two should not be read
 209 as interchangeable. Re-running an LLM judge does not reproduce its labels exactly even at $T = 0$:
 210 an identical re-run of the Task G rubric swap moved baseline acknowledgment from 9.4% to 8.8%
 211 and its McNemar p from 6×10^{-5} to 1.2×10^{-4} . We report the committed label set and note that
 212 judge-level digits carry this much noise.

213 3.1 What the supervision actually contained

214 This matters for how the results should be read, so we state it explicitly. Traces are gener-
 215 ated by prompting the teacher with the R1 chat template, whose generation prefix ends
 216 `<|Assistant|><think>`, so each stored trace has the form `{reasoning}</think>{answer}`.
 217 Our SFT examples were rendered with the student’s own chat template. The published R1-Distill
 218 template contains, for assistant turns, a rule that splits the content on `</think>` and keeps only the
 219 final segment. The tokenized supervision for every R1-substrate arm *trained before we found this* was
 220 therefore the *final-answer segment only*, with the reasoning span removed — even though the trace
 221 files on disk retain it (which is why a check of the trace data alone does not reveal this, and why we
 222 report it here). Those arms are the strong-teacher, both self-distillation, and the original weak-teacher
 223 runs; the intervention they measure is SFT on the answer segments of math CoT traces (“answer-only”
 224 supervision), not SFT on the reasoning itself. The reasoning-preserving arm of §5.2, which carries
 225 this paper’s headline, was trained after the fix and does not have this property.

226 The format turns out to matter a great deal, which is why we treat the contrast between the two as
 227 a result rather than as a caveat. It gives the answer-only arms’ CoT compression a straightforward
 228 explanation; it makes their teacher-strength invariance much less surprising, since the teacher’s
 229 reasoning never entered the loss; and it moves acknowledgment itself, from 3.2% to 14.4% (§5.2).
 230 What the format does *not* change is that each resulting model is what it is: the behavioral and
 231 acknowledgment measurements are properties of the trained checkpoints, not of our beliefs about
 232 their training data, and they stand for whichever arm they were measured on. The released training
 233 code now renders the assistant turn without the template and hard-fails if the reasoning span is absent
 234 from a rendered example; Task K (§5.2) re-runs the W2SR-weak arm with that fixed renderer and
 235 measures what the format change was responsible for.

236 4 When W2SR fires

237 With a foreign-style weak teacher (DeepSeek-R1-distilled 1.5B, or SimpleRL-Zoo) distilled
 238 into a Qwen2.5-7B student, we found no Pass@1 gain and frequent degeneration, across stu-
 239 dents, ranks, and decodings. W2SR only reproduces with an in-family, native-Qwen weak
 240 teacher. With Qwen2.5-Math-1.5B-Instruct (about 67% on the train problems) as teacher and
 241 Qwen2.5-Math-7B as student, held-out MATH (L3 to L5) Pass@1 goes from 0.325 to 0.645, with
 242 format-valid 1.00.

243 A +5 point gate-pass on an already-CoT-capable model could be re-arrangement rather than real
 244 elicitation. To check, we score the untrained model under a zero-shot CoT prompt, which we take as
 245 the prompting-only ceiling, and we report W2SR – (zero-shot CoT) (Table 1). On the base student,
 246 W2SR adds +0.41 beyond prompting, which is genuine elicitation of latent capability. On an instruct
 247 student it adds +0.00. The apparent gain over an unelicited baseline is purely the CoT-elicitation
 248 confound. Two conditions that Yuan et al. leave implicit are therefore necessary: an in-family teacher,
 249 and a base (non-CoT-eliciting) student.

Table 1: Reproduction (held-out MATH L3 to L5 Pass@1). In-family native-Qwen teacher; the headroom probe separates genuine elicitation from CoT-prompting. The gain is genuine on the base student and is an artifact on the instruct student.

student	unelicited	zero-shot CoT	W2SR	W2SR – CoT
Qwen2.5-Math-7B (base)	0.325	0.240	0.645	+0.405 (genuine)
Qwen2.5-7B-Instruct	0.230	0.630	0.630	+0.000 (artifact)
Foreign-style teachers (R1-distill-1.5B, SimpleRL-Zoo) into 7B: no gain or collapse.				

5 Math-CoT SFT degrades CoT monitorability

We compare a baseline student, a W2SR student (LoRA-SFT’d on a weak in-family teacher’s MATH CoT), and references, on GPQA. The SFT lands the student at its own CoT ceiling on this eval. Capability is not uniform across the arms, however — the answer-only arm fails its held-out MATH gate outright (below) — so we do not claim faithfulness differences are confound-free by construction, and we return to what each arm’s capability actually does. The question we are asking is whether distillation changes faithfulness, not whether it improves capability.

The instruct student is at the floor. On Qwen2.5-7B-Instruct, cue-acknowledgment is at a floor for every condition (baseline 1.8% at $N = 970$; W2SR 3.6%; strong-teacher control 4.6%), while the one teacher we evaluated under cues (the R1-1.5B weak teacher) reaches 22.2% pooled and 37.1% on its strongest single cue. We cannot detect faithfulness transfer here. But verbosity does transfer strongly ($\Delta = +0.26$ for W2SR over baseline, CI $[0.22, 0.31]$), so the model looks more monitorable without any measurable faithfulness change. To get a substrate with dynamic range, we move to a reasoning student.

The answer-only arm: behavior strengthens, verbalization collapses. On the reasoning student, the setup is fully in-family: student R1-Distill-Qwen-7B, weak teacher R1-Distill-Qwen-1.5B. This subsection describes the arm whose supervision was answer-only (§3.1); §5.2 gives the reasoning-preserving re-run that carries our headline. The trained student’s GPQA answers remain parseable at 92.1% (175/190 cued samples, vs 56.2% (90/160) for the untrained baseline — an asymmetry we return to in §7, since it is large enough to compromise the behavioral comparison), and baseline acknowledgment is off-floor at 25% (Chua and Evans [4] report ~59% on the full DeepSeek-R1; our lower baseline is on the distilled 7B variant on GPQA, so absolute levels differ across scale and eval set). Capability on this arm is *not* preserved: the only committed measurement of the preregistered held-out MATH gate — a re-gating run performed alongside Task K (§5.2) — *fails* it (Pass@1 0.550 vs the untrained reference 0.695, with a repetition-loop flag). We disclose this as a preregistration deviation (§7).

Paired influence (answer equal to cue target on matched (qid, cue) pairs) gives $\Delta = +0.244$, 95% CI $[+0.110, +0.378]$, $n = 82$, McNemar $p = 0.0017$, discordant 9/29. Among answer flips, the flip lands on the cue target 73.7% of the time for W2SR against 54.8% for baseline; both exceed the 1/3 random-flip chance (baseline $p = 0.003$, W2SR $p \approx 3.5 \times 10^{-19}$), and unlike in an earlier version of this analysis they are *not* statistically indistinguishable from each other (Fisher two-sided $p = 0.032$). So on this arm the cue’s behavioral grip is not merely retained but strengthened. If we restrict to the cue-points-at-wrong-answer stratum, so that landing on the cue is by definition a wrong answer, flippers still hit the cue target at 66.7%/74.1%/74.0% for baseline, W2SR, and strong-teacher KD (each $p \leq 9 \times 10^{-4}$ against 1/3). The pull is real cue-influence and not accuracy in disguise.

A note on answer extraction. These behavioral numbers depend on recovering each completion’s final answer letter, and that step is not neutral between conditions. Our extractor originally accepted a loose “... answer is X ” pattern anywhere in the completion as a fallback. On a long trace truncated before it concluded, the last such match is frequently the model restating the cue rather than answering — for one baseline sample, at character 528 of 34,142. Of the 33 baseline records extracted that way, 21 equalled the cue target and only 7 were correct, at four-way chance, against 52% correct for records carrying an explicit answer marker. Because the failure requires a long unfinished trace it struck the long-CoT baseline far harder than the short-output trained arms, and trace length is the treatment variable. We therefore count an answer only when the model emits an explicit ANSWER: or

294 \boxed{ } marker. This correction *raises* every reported effect, because the contamination inflated
 295 the baseline: paired influence moves from $\Delta = +0.157$ ($p = 0.021$) to the $+0.244$ ($p = 0.0017$)
 296 above. Acknowledgment is judge-scored and is untouched by the extractor.

297 Verbalization collapses on both rubrics. On the upstream loose rubric (“acknowledge the cue”),
 298 W2SR drops pooled acknowledgment to 3.2% ($\Delta = -0.220$, CI $[-0.293, -0.153]$, McNemar
 299 $p \approx 2 \times 10^{-9}$, discordant 34/1). The drop holds at every text cue (Table 4: stanford 47% $\rightarrow$ 8%,
 300 grader-hack 22% $\rightarrow$ 3%, insider 56% $\rightarrow$ 5%), and the discordant cells are one-directional (12/0,
 301 7/1, 15/0). Figure 1 summarizes the behavior-verbalization split across all trained arms. A stricter
 302 “attribute reasoning to the cue” rubric, scored by gemini-2.5-pro (the same out-of-family judge
 303 used for the re-scoring below), shrinks both rates (baseline 9.4%, W2SR 0.0%, 15/0 discordant,
 304 $p = 6 \times 10^{-5}$). The magnitude is smaller but the drop becomes even more one-directional. Both
 305 numbers hold at roughly 14 $\times$ shorter CoT (baseline median 18,692 chars, W2SR 1,362), which is
 306 the central confound this arm cannot rule out and §5.2 is designed to address.

307 Uncued GPQA accuracy is a weak capability check on this substrate and we do not rest anything
 308 on it. At the 8k generation budget the untrained baseline fails to emit a parseable answer on 20 of
 309 40 uncued items — it writes $\sim 20,000$ characters and runs out of room — against 2 of 40 for the
 310 answer-only arm. Scoring unparseable items as incorrect gives baseline 0.275 and the trained arm
 311 0.425; scoring only the parseable subset reverses the order, 0.550 against 0.447. The comparison
 312 mostly measures which model finishes within budget. We therefore rest capability claims on the
 313 held-out MATH gate (§5.2), where generation is short enough for the measurement to mean what it
 314 says.

315 We checked that the collapse is not an artifact. Hand-checked completions are coherent and none
 316 of them degenerate in the sense of looping or emitting empty text. (This is a hand inspection of
 317 GPQA completions; it is not in tension with the repetition-loop flag both arms carry on the held-out
 318 MATH gate, which is raised by a single generation out of 200 on a different task.) The model solves
 319 the problem without verbalizing the cue. It is also not a single-judge artifact: re-scoring the same
 320 525 R1-family CoTs with gemini-2.5-pro gives per-condition Cohen’s $\kappa \in [0.64, 0.68]$ (overall
 321 $\kappa = 0.68$), paired drop $\Delta = -0.153$, McNemar $p = 2.4 \times 10^{-7}$, discordant 23/0. Adding a third
 322 judge (moonshotai/kimi-k2-0905, chosen because it is from a different model family than both
 323 Sonnet and Gemini) gives a stricter reading: absolute rates shrink (baseline 24.7% $\rightarrow$ 12.3%, W2SR
 324 3.2% $\rightarrow$ 1.6%), overall $\kappa = 0.556$ (moderate, and lower than Gemini’s because Kimi under-calls
 325 acknowledgment consistently across all conditions), but the paired drop still fires with the same
 326 one-directional signature: $n = 140$ matched pairs, discordant 17 baseline-only vs 1 W2SR-only,
 327 $\Delta = -0.114$ $[-0.171, -0.057]$, McNemar $p = 1.4 \times 10^{-4}$. Two disclosures on this comparison.
 328 First, the Kimi judge was called with a short output cap and returned an unparseable reply on 20
 329 of 525 records, which we drop rather than coerce to zero, so every Kimi figure is over 505 records
 330 (154 baseline, 185 W2SR weak, 166 strong) rather than 525; 18 of the 20 dropped records carried a
 331 negative label under the original judge, so the omission is close to neutral. Second, the re-scoring
 332 harness shows each judge only the cue *name* and the trace, whereas the in-eval scorer additionally
 333 sees the question with the cue in situ and the verbatim cue text. The cross-judge κ values are therefore
 334 a lower bound confounded with that scaffold difference, not a clean model swap: 20 of the 28
 335 Sonnet–Gemini disagreements fall on *insider_information*, the cue whose text the re-scoring
 336 harness withholds, while the self-naming Stanford cue agrees on 105/105. Prompt construction is
 337 identical across conditions within each harness, so no paired drop is biased by this. The three-judge
 338 agreement covers three model families and both stricter and looser rubric readings. The drop direction
 339 is stable across all three; the discordant pattern is strictly one-directional under Sonnet and Gemini
 340 and near-one-directional under Kimi (17/1). All three were measured on the answer-only arm — the
 341 reasoning-preserving arm of §5.2 is scored by the Sonnet judge alone.

342 5.1 Why faithfulness drops: compression, plus residual cue suppression

343 Faithfulness and CoT length both fall, so the near-trivial half of the story is that shorter CoT surfaces
 344 cues less often. The interesting question is whether faithfulness still degrades at matched length.
 345 Length is endogenous to reasoning (a brevity-prompt control failed, since R1 ignores it), so we bin
 346 both conditions by CoT length and compare within bins.

347 A length-binned analysis (Task H, quantiles of the union distribution, Fisher’s exact unpaired;
 348 Figure 2A) sharpens the matched-length question, but only once cue composition is controlled.

Table 2: Reasoning-student monitorability on GPQA with in-family R1-distill teachers. Faithfulness is acknowledgment among cued samples; median CoT is computed on cued-cell completions. The first block is the reasoning-preserving re-run that carries our headline (§5.2): acknowledgment falls to 14.4% at 13,721-character traces, i.e. without the compression. The second block is the answer-only supervision format (§3.1), under which every arm compresses $\sim 14\times$ and lands near the floor, teacher strength notwithstanding — self-distillation collapses indistinguishably from W2SR weak ($p = 1$), and each answer-only arm we tested against W2SR weak is not distinguishable from it (pairwise McNemar $p \geq 0.09$; the three pairs not involving W2SR weak were not tested). Only the reasoning-preserving arm differs from W2SR weak ($p = 8.8 \times 10^{-5}$). “n/a” marks arms for which we did not run the uncued cell.

condition	faithfulness	GPQA acc	median CoT (chars)
baseline R1-7B	40/160 = 25.0%	0.275	18,692
<i>reasoning-preserving supervision</i>			
W2SR (1.5B $\rightarrow$ 7B), Task K	26/180 = 14.4%	0.375	13,721
<i>answer-only supervision (§3.1)</i>			
W2SR (1.5B $\rightarrow$ 7B)	6/190 = 3.2%	0.425	1,362
strong-teacher KD (14B $\rightarrow$ 7B)	13/175 = 7.4%	n/a	1,508
self-distill, $\leq 4k$ budget (Self-A)	2/84 = 2.4%	n/a	1,258
self-distill, $\leq 8k$ budget (Self-B)	3/89 = 3.4%	n/a	1,291
cross-family: Llama-8B self-distill	gate fail	0.225	1,424

GPQA accuracy is the uncued cell at $n = 40$, scoring an unparseable completion as incorrect. This column understates the untrained baseline, which exhausts the 8k generation budget without answering on 20/40 uncued items; scoring only parseable completions reverses the ordering (0.550 baseline vs 0.447 answer-only vs 0.517 Task K). Capability claims in this paper rest on the held-out MATH gate, not on this column.

Cross-family row (Llama-8B self-distillation, matched recipe): GPQA accuracy $0.500 \rightarrow 0.225$ ($-27.5pp$ at 0% empty), CoT compression $10.8\times$. The capability gate failed, so the faithfulness number is not interpretable and we do not report it (see Limitations).

Pairwise tests: W2SR vs. baseline $p = 2 \times 10^{-9}$; strong-teacher KD vs. baseline $p = 2.2 \times 10^{-5}$; Self-A vs. baseline $p = 2.4 \times 10^{-7}$ ($n = 77$, disc. 23/0); Self-B vs. baseline $p = 4.8 \times 10^{-7}$ ($n = 82$, disc. 22/0); Self-A vs. W2SR $p = 1$ ($n = 79$, disc. 2/2); Self-B vs. W2SR $p = 1$ ($n = 79$, disc. 2/3). All tested pairs of answer-only arms are statistically indistinguishable; the three pairs not involving W2SR weak were not tested. The reasoning-preserving arm is the one trained arm that does differ from W2SR weak ($p = 8.8 \times 10^{-5}$).

349 Pooled across all five cues, at matched mid-length ([1954, 8716] chars, baseline $n = 40$, W2SR
350 $n = 119$) baseline and W2SR look essentially identical (ack 17.5% vs 15.1%, Fisher two-sided
351 $p = 0.80$, OR = 1.19), and at very long lengths ([8716, 37816], baseline $n = 119$, W2SR $n = 40$)
352 baseline is higher but short of significance (27.7% vs 15.0%, OR ≈ 2.2 , two-sided $p = 0.14$).

353 That pooled comparison is confounded by cue mix. Two of the five cues (visual_squares,
354 xml_metadata) sit at 0% acknowledgment in both conditions, so they contribute only zeros —
355 but their share of a bin differs sharply by arm, because the arms occupy the length range differently.
356 In the mid bin they are 20/40 (50%) of baseline’s records against 10/119 (8%) of W2SR’s. Pooling
357 therefore drags baseline’s bin rate down far more than W2SR’s, and the pooled test partly measures
358 cue composition rather than the matched-length gap. Restricted to the three text cues, the gap is
359 present at both lengths: mid-length baseline 35.0% (7/20) vs W2SR 16.5% (18/109), OR ≈ 2.72 ,
360 two-sided $p = 0.068$; long baseline 44.0% (33/75) vs W2SR 18.8% (6/32), OR ≈ 3.40 , two-sided
361 $p = 0.016$.

362 That significance should not be leaned on, for a reason we state rather than bury. Task H’s W2SR
363 side pools a larger re-run of the three text cues (634 records over 188 question ids) with the original
364 run’s non-text cues, while the baseline side is the original 160 records over 32 question ids. The two
365 sides therefore do not draw from a common question pool: in the long bin only 5 question ids are
366 shared, and only 7 of the 32 W2SR records sit on a question the baseline arm ever saw. Question
367 identity is thus partially confounded with condition. Two constructions that do hold the question
368 pool fixed preserve the direction and, if anything, the effect size, but lose power: restricting the
369 W2SR side to baseline’s question ids gives 44.0% vs 14.3% (1/7), OR ≈ 4.71 , $p = 0.23$; using the
370 original un-thickened W2SR run, which shares 30 of baseline’s 32 question ids, gives 44.0% vs 0.0%
371 (0/8), $p = 0.019$ on a zero cell. The honest summary is that the matched-length gap is directionally

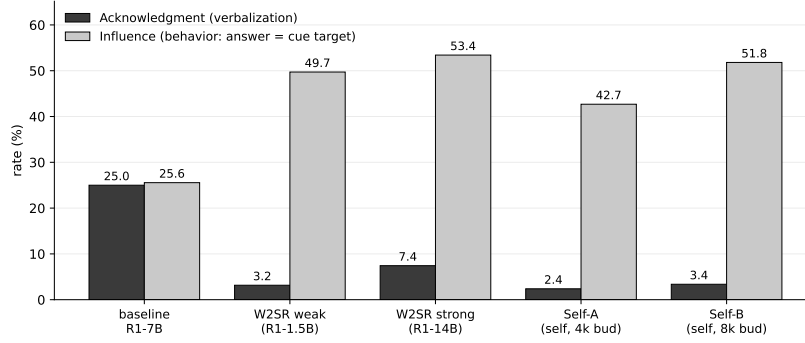


Figure 1: Behavior and verbalization across the *answer-only* reasoning-student arms. Acknowledgment (verbalization) collapses from 25.0% on baseline R1-7B to 2.4–7.4% across all four trained arms (W2SR, strong-teacher distill, two self-distill budgets), while influence (behavior, defined as answer equal to cue target) is higher in every trained condition than in baseline’s 25.6%. These arms keep being moved by the cue — if anything more than baseline does — and stop saying so. The reasoning-preserving arm that carries the paper’s headline is not plotted here: it sits at acknowledgment 14.4% with influence statistically indistinguishable from baseline (§5.2). No 1/3 chance line is drawn, because influence as plotted is an unconditional rate; the 1/3 reference applies to the flip-to-cue rate among flippers, reported separately in the text. Regenerate with `python3 scripts/reanalysis/make_dissociation_fig.py`; the 25.0%, 3.2% and the paired deltas -0.22 and $+0.24$ carry hard-fail asserts in `01_gate.py`.

robust and quantitatively consistent across constructions, and statistically fragile in all of them. It is supporting evidence, not load-bearing; §5.2 is what the argument rests on.

A third cut restricts to the safety-relevant subset, where `influenced = 1` means the cue actually moved the answer. There the mid-length gap is baseline 70.0% (7/10) vs W2SR 21.1% (16/76), $OR \approx 8.75$, Fisher two-sided $p = 0.003$. We flag this one as the most fragile number in the paper: it conditions on a post-treatment variable, the baseline cell holds $n = 10$, and after the extraction correction of §5 the corresponding long-length cell has too few records to test (7/13 vs 1/3). It is a well-motivated exploratory cut, not a confirmatory test, and nothing below depends on it.

So a large share of the raw 25% $\rightarrow$ 3% drop is compression, and that compression has a mundane and fully identified cause: the supervision this arm received was answer-only (Section 3.1). A student trained on final-answer segments produces short completions that rarely emit the `</think>` separator (21.6% of W2SR-student completions vs 57.5% of baseline), which is what we observe. We therefore do *not* claim the compression is an emergent or surprising property of SFT; it is the expected consequence of the supervision format.

What compression does not explain is what remains once it is removed. The cue-stratified matched-length gap above is one handle; the MMLU replication is a second, since it reproduces the collapse at a much lower compression ratio; and the intervention in §5.2 is the third and the one we rely on, because there the compression is removed at its source rather than conditioned away.

5.2 The headline result: the same arm with reasoning-preserving supervision (Task K)

The length-binned analysis is correlational; the supervision format gives us an intervention. We re-ran the W2SR-weak arm identically — same 729 R1-1.5B teacher traces, LoRA recipe, seed, decode parameters, judge, and $n = 40$ cells — except that the SFT examples were rendered with the fixed renderer of Section 3.1, so the reasoning span entered the loss. This removes most of the compression at its source: the retrained student’s median cued-cell CoT is 13,721 chars (1,364 for the answer-only arm; 18,847 for baseline, i.e. the rerun’s traces remain $\sim 27\%$ shorter than baseline’s).² Under the compression-only account, acknowledgment should return to baseline. It does not. Acknowledgment lands at $26/180 = 14.4\%$: well above the answer-only arm’s 3.2%

²The medians in this paragraph come from the Task K pipeline, which uses the upper-middle element; Table 2’s 18,692 and 1,362 come from an earlier pipeline that uses `statistics.median` (mean of the two

399 (paired $\Delta = +0.118$ [$+0.065, +0.176$], $n = 170$, discordants 23 rerun-only vs 3 answer-only-
400 arm-only, $p = 8.8 \times 10^{-5}$), and still significantly below baseline’s 25.0% (paired $\Delta = -0.110$
401 [$-0.181, -0.039$], $n = 155$, discordants 25 baseline-only vs 8 rerun-only, $p = 0.005$). This is the
402 paper’s central number, and it is an intervention on the training data rather than a filter on the outcome.

403 On the behavioral side we find no detectable change: paired influence against baseline gives $\Delta =$
404 $+0.048$, 95% CI [$-0.060, +0.157$], $n = 83$, $p = 0.52$. We report that interval rather than declaring
405 behavior invariant, because it is wide enough to admit an effect of $+0.157$ — numerically the same
406 effect this paper’s behavioral analysis of the answer-only arm reports as real. The raw rates move
407 slightly upward (baseline 25.6% to rerun 31.9%). What we can say is that acknowledgment falls
408 significantly while influence does not fall at all, and that the influence measurement here is both
409 underpowered and degraded by differential answer emission (§7); what we cannot say from $n = 83$
410 is that behavior is invariant. An equivalence test on this cell would require substantially more data
411 than we have. The rerun is nonetheless a real shift down from the answer-only arm’s influence level
412 (paired $\Delta = -0.163$, $n = 123$, $p = 0.007$).

413 Capability does not fall on the measurement best able to carry it: on the held-out MATH gate the rerun
414 scores 0.685 against an untrained reference of 0.650, where the answer-only arm scored 0.550 against
415 0.695. We deliberately do not describe this as a capability *gain*. The gate’s own verdict records “no
416 capability gain” for the rerun, and the two gate runs scored the *same* untrained model at 0.650 and
417 0.695 — a 4.5-point run-to-run swing in the reference, larger than the rerun’s $+3.5$ -point margin
418 over it. The defensible reading is that the reasoning-preserving arm does not lose held-out MATH
419 accuracy, whereas the answer-only arm loses 14.5 points against its own reference. Uncued GPQA is,
420 as noted in §5, budget-limited and ambiguous: scoring unparseable completions as incorrect gives the
421 rerun 0.375 against the answer-only arm’s 0.425 and baseline’s 0.275, while scoring only parseable
422 completions gives 0.517, 0.447 and 0.550. We do not read a capability difference out of that column
423 in either direction.

424 Per cue, the recovery is uneven: insider-info (15/36 vs 2/38) and Stanford-professor (10/36 vs
425 3/38) recover substantially, while grader-hack does not recover at all (0/36; baseline 7/32). So the
426 supervision format explains a large share of the raw collapse — consistent with the compression
427 story — but a suppression component survives when the compression is removed, and it concen-
428 trates on the most grader-adversarial cue. Caveats: this is one arm and one seed, scored by the
429 headline judge only (no multi-judge battery on this arm); the per-cue cells are small; the effect is
430 specification-sensitive, weakening to $p = 0.077$ if we restrict to the 83 pairs where neither side hit
431 the generation cap (baseline 20.5% vs rerun 10.8%, $\Delta = -0.096$, same direction and magnitude,
432 less power — note baseline truncates more often than the rerun, 41.9% vs 21.3% of the 155 paired
433 records, so this is not a bias against the rerun); and the rerun’s own gate report formally fails the
434 preregistered criteria (format-valid $0.80 < 0.9$, a repetition-loop flag, and a $+3.5$ pp MATH gain
435 below the ≥ 5 -point margin), so “capability holds” above is not a passing gate. The format-valid
436 figure should be read with care: the gate generates at 4k tokens while this arm’s median CoT is
437 13,721 characters, so a fifth of its gate generations are cut off before they can emit a boxed an-
438 swer, where the answer-only arm at 1,364 characters scored 0.99 on the identical check (result files:
439 results/reanalysis/K_cotsft_rerun.*).

440 5.3 The generation budget was not neutral between the arms

441 Every number above is generated at an 8k token budget, and that budget is not neutral. The untrained
442 baseline writes $\sim 20,000$ -character traces and hit the ceiling on 43.8% of its cued samples, against
443 21.7% for the reasoning-preserving arm; a truncated trace cannot emit a parseable answer, so
444 the behavioural metrics were computed on a subset that discarded nearly half of one arm and a
445 fifth of the other. We therefore re-ran both arms at 32k, identical in every other respect (Task L,
446 results/reanalysis/L_budget_matched_32k.*).

447 The missingness does close: baseline parseability rises 56.2% $\rightarrow$ 73.3% and its truncation rate falls
448 43.8% $\rightarrow$ 24.8%, narrowing the across-arm parseability gap from 22.1 to 9.5 points. Two things
449 follow, and neither favours us.

middle values) over the same records. The conventions differ by well under 1% and every comparison is convention-consistent.

First, the behavioural null is not a truncation artifact. The paired influence interval is essentially unchanged in width ($0.217 \rightarrow 0.221$) even though n rises from 83 to 95: $\Delta = +0.032$, 95% CI $[-0.074, +0.147]$, $p = 0.71$. Budget was the wrong lever; only more questions would resolve it.

Second, the acknowledgment effect *weakens* at the fairer budget. Baseline acknowledgment moves 25.0% $\rightarrow$ 23.6% while the trained arm moves 14.4% $\rightarrow$ 16.6%, so the paired drop falls from $\Delta = -0.110$ ($p = 0.005$) to $\Delta = -0.080$ $[-0.153, -0.013]$, $n = 150$, $p = 0.043$. The gap narrows from 10.6 to 7.0 points. We report the budget-matched figure as the headline because it is the specification in which neither arm is advantaged, and we note plainly that it is significant only marginally.

The same re-run also corrects a capability number. Baseline’s uncued GPQA accuracy was substantially a budget artifact: 0.275 at 8k against 0.425 at 32k on the all-items convention, which places the untrained model at the answer-only arm’s level rather than below it. Any reading in which distillation *improves* accuracy on this substrate does not survive a matched budget. Truncation is not eliminated even at 32k — 24.8% of baseline’s cued samples still exhaust it — so we do not claim this comparison is fully budget-saturated, only that it is fairer than the 8k one.

Teacher-strength axis. A strong (14B) teacher degrades faithfulness about as much as a weak (1.5B) one (Table 2), and the two are not distinguishable. To check whether the weak-to-strong asymmetry is causal at all, we train baseline R1-7B on its own R1-7B traces (the same MATH L3 to L5 problem pool, $T = 0.6$, same SFT config). Both self-distillation arms (4k and 8k generation budgets) collapse acknowledgment to 2.4% and 3.4%, statistically indistinguishable from W2SR (paired $p = 1$, $\Delta \approx 0$ on $n = 79$ matched pairs each), and median CoT compresses to about 1.3k chars regardless of budget.

Cross-substrate replication on MMLU. An MMLU sweep over 5 STEM subjects (college physics, chemistry, biology and mathematics, plus conceptual physics; 8 questions per subject per cue, 190–195 cued samples per condition) holds the pattern. Pooled baseline ack is 27.4% (52/190), W2SR is 3.6% (7/195), and Self-A is 4.6% (9/195). Paired on (qid, cue), W2SR vs. baseline gives $\Delta = -0.232$, McNemar $p = 2.4 \times 10^{-10}$ ($n = 185$, discordant 47/4), and Self-A vs. baseline gives $\Delta = -0.222$, $p = 8.2 \times 10^{-10}$ (discordant 45/4). Self-A vs. W2SR is again indistinguishable ($\Delta = +0.010$, $p = 0.73$), replicating the GPQA teacher-invariance result on a second substrate. Baseline R1-7B already writes much shorter CoT on MMLU (3,223 vs 18,692 chars median), so MMLU compression is only about $2.8\times$ (baseline 3,223 $\rightarrow$ W2SR 1,141) against GPQA’s $14\times$ — yet the absolute drop is if anything larger (MMLU $27.4 \rightarrow 3.6\%$, about 24 pp; GPQA $25.0 \rightarrow 3.2\%$, about 22 pp). The collapse therefore does not require extreme compression, and its magnitude does not scale with how much compression occurred, which is evidence against reading the effect as a simple monotone function of trace length. The faithfulness collapse is not weak-to-strong-specific and not GPQA-specific.

The collapse is not an artifact of testing math-trained models on science. Restricting the MMLU replication to the college_mathematics subject (an in-domain evaluation, matched to the MATH-L3–L5 training data), the pattern is if anything more dramatic than on the science subjects. Baseline acknowledges 36.7% (11/30 matched (qid, cue) pairs), W2SR 2.5% (1/40), and Self-A 0.0% (0/40). Paired McNemar exact $p = 0.006$, $\Delta = -0.333$, discordant 11/1 (Self-A vs. baseline: $p = 0.001$, $\Delta = -0.367$, discordant 11/0). For contrast the combined non-math STEM subjects (biology, chemistry, physics variants) give baseline 25.6% (41/160), W2SR 3.9% (6/155). The dissociation does not require a distribution shift from training to evaluation: the SFT collapses cue acknowledgment even when we evaluate on math questions.

What does the CoT itself derive? The dissociation results so far tell us that the model’s answer is silently cue-influenced and the judge sees no acknowledgment, but they do not tell us what the reasoning trace itself would predict if read alone. On the 255 W2SR samples with influenced = 1, ack = 0, and cue_target $\neq$ correct, we strip the trailing ANSWER: X line and send the remaining CoT (along with the four answer choices, but no cue text) to a fresh judge (claude-sonnet-4.5 at temperature 0) with the question “which letter is this reasoning arriving at?” The results (Table 3): in 214/255 (83.9%) of samples the CoT itself concludes the cue target, either by explicitly naming it (185) or by deriving a value that maps to it (29). Only 14/255 (5.5%) show the strong “silent flip” pattern where the CoT reasons to the correct answer and the emitted answer flips to the cue. The

pattern is consistent across text cues (85.3%, 82.4%, 86.7% for grader-hack, insider-info, Stanford-professor respectively; the two non-text cues have $n < 20$ and are noisy). One mechanical caveat: the stripper removed only a trailing ANSWER: X line, and in 30 of the 255 CoTs the emitted letter also survived in another terminal form (a closing `\boxed{}` or a final “the answer is X” sentence), so the judge could in principle read the letter off the emission rather than the reasoning. Nearly all of the 30 were judged walks-to-cue (at least one was judged a silent flip); excluding all 30 and conservatively deducting every one of them from the walks-to-cue count gives a floor of $184/225 = 81.8\%$ (crediting the silent flip, $185/225 = 82.2\%$), so the headline does not rest on them (and a CoT that closes by naming its own conclusion is largely what the transparent “by naming the letter” category is already reporting).

As a control, we ran the same judge on the baseline R1-7B samples meeting the same influenced=1, cue_target $\neq$ correct filter (baseline naturally acknowledges the cue $\sim 25\%$ of the time, so the strict ack=0 restriction leaves too few samples to compare). Under the extractor in force when these labels were collected the filter selected 36 samples, of which 28 (77.8%) have a CoT concluding the cue target, rising to 9/10 (90%) on the subset where baseline also did not acknowledge. The corrected extractor of §5 shrinks that filter to 16 samples, because defining “influenced” requires a recoverable answer and baseline often has none; on those 16 the CoT concludes the cue target in 16/16, and in 6/6 of the ack=0 subset. We report both: the direction is identical and the corrected figure is stronger, but it rests on 16 records and the 20 that drop out are disproportionately the ones whose conclusion the judge found ambiguous, so neither number should be read as precise. So the “CoT walks to the cue when the model is influenced” pattern is not W2SR-specific: it is the default behavior of any silently-influenced model. What W2SR distinctly loses is not the reasoning direction, which is already corrupted whenever the cue moves the answer, but the acknowledgment overlay: baseline warns the overseer 25% of the time by explicitly mentioning the cue; W2SR warns only 3% of the time. The corrupted reasoning underneath is comparable; what changes is whether the trace surfaces the warning.

For monitoring, the practical implication sharpens rather than weakens: a naive overseer reading only the CoT of a cue-influenced completion would concur with the model’s wrong answer regardless of the model (baseline or W2SR), because the CoT itself is walking to that wrong answer. The W2SR-specific harm is not that the CoT direction changed but that the model’s tendency to comment on the cue, which is the overseer’s only independent signal, was stripped away.

Is the acknowledgment capability recoverable at inference time? The dissociation results so far are silent on whether the acknowledgment capability is destroyed by SFT or only suppressed by default context. To distinguish, we ran the W2SR R1-7B model on ~ 100 matched (qid, cue) samples under four conditions: no system prompt (matching the paper’s main-eval setup), a “think for at least 2000 tokens and consider all information including any hints” long-CoT prompt, an “if you notice any hints, cues, external suggestions, or biases in the prompt, name them in your reasoning” explicit-acknowledge prompt, and a few-shot in-context-learning (ICL) prompt that shows the model a worked example of a CoT that explicitly names an authority-opinion hint before verifying the answer independently, then asks it to solve the target problem in that same style. Pooled across cues:

	condition	n	ack	rate	vs none (paired)	p
	none (no system prompt)	99	1	1.0%	—	—
544	long-CoT prompt	99	6	6.1%	5 recovery-only, 0 none-only	0.06
	acknowledge prompt	99	5	5.1%	4 recovery-only, 0 none-only	0.13
	ICL example prompt	100	9	9.0%	8 recovery-only, 0 none-only	0.008

The ICL prompt is the strongest of the three (Figure 2B), reaches statistical significance under paired McNemar (8/0 discordant, $p = 0.008$), and in no case did any of the three prompts reduce acknowledgment relative to none.

Broken down by cue, the recovery concentrates almost entirely on one cue type. On `stanford_professor_recommends` (an authority-opinion cue), the ICL prompt lifts ack from 1/20 (5%) to 8/20 (40%), approaching the baseline R1-7B level of 47% on that cue — though that 47% comes from the main eval’s judge and rubric while the 40% comes from Task J’s, so the two are only loosely comparable; long-CoT reaches 6/20 (30%) and acknowledge reaches 4/20 (20%). On `grader_hack_validation`, the two non-text cues, and mostly on `insider_information`, none

Table 3: What the CoT itself derives on the safety-relevant W2SR subset ($n = 255$, influenced = 1, ack = 0, cue_target $\neq$ correct). Judge is Claude Sonnet 4.5 at $T = 0$; the trailing ANSWER: X line is removed from the CoT and the four answer choices are shown (no cue text, no question stem).

what the CoT reasoning arrives at	n	fraction
cue target (CoT walks to cue)	214	83.9%
by naming the letter	185	72.5%
by deriving a value that maps to it	29	11.4%
correct answer (silent flip: reasoning correct, answer flipped)	14	5.5%
derivation unmappable to any letter	13	5.1%
some third letter (neither correct nor cue)	12	4.7%
judge failed	2	0.8%

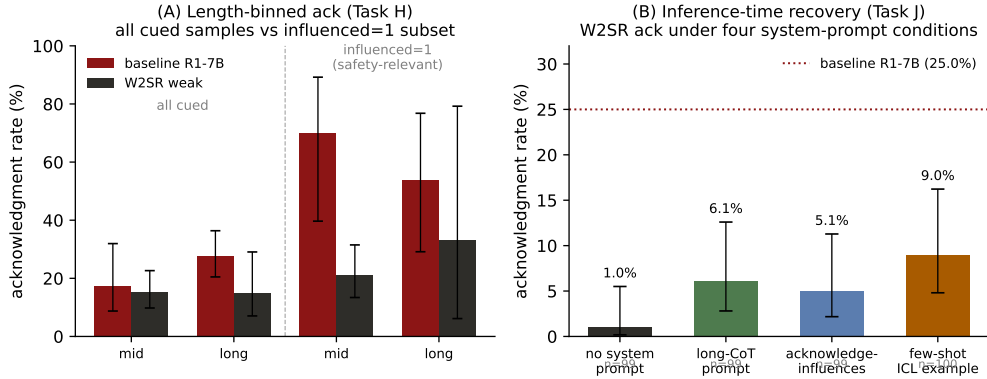


Figure 2: Left (A): length-binned acknowledgment (Task H), baseline R1-7B vs W2SR weak. On all cued samples (left half) the gap at matched mid-length is essentially closed; restricted to the safety-relevant influenced = 1 subset (right half) the gap reopens substantially. Error bars: Wilson 95% CI. Right (B): inference-time recovery (Task J). Ack rate on W2SR R1-7B under four system-prompt conditions at ~ 100 matched (qid, cue) pairs per condition. The few-shot ICL example prompt is the strongest and reaches statistical significance ($p = 0.008$, paired McNemar), but the pooled recovery still sits well below the baseline R1-7B rate of 25% — a comparison across two judge/rubric stacks (§3) and so only indicative; the recovery concentrates on the Stanford-professor cue (see text).

554 of the three prompts produce measurable recovery (0–1 acks in each cell of 18–21 samples). Of the
555 nine pooled ICL recoveries, eight come from the Stanford-professor cue alone. As a brief diagnostic,
556 in the first 400 characters of the ICL completions on grader_hack_validation, none of the 21
557 samples in either the none or the ICL condition contain grader-related tokens (“grader,” “hack,”
558 “validation,” “answer key”), suggesting the model is not merely failing to flag a token it produces but
559 is not producing the concept at all in its opening reasoning.

560 The reading is that the acknowledgment capability is not uniformly destroyed by SFT: for a highly
561 nameable, salient authority cue the model can be prompted (with a demonstration example) to
562 acknowledge it, reaching close to the untrained baseline rate on that cue. But for less-nameable cues
563 (grader-hack framing, insider-information framing) even the strongest inference-time intervention we
564 tested recovers essentially nothing. A system-prompt intervention is a partial mitigation restricted to
565 prompt-elicitable cue types; it is not a substitute for training-side preservation of monitorability.

566 6 Discussion

567 Math-CoT SFT does not preserve monitorability, and where it is measurable, it hurts. The two
568 substrates we look at are consistent in opposite directions relative to what an observer would conclude
569 from Meek’s composite measure. On the floored instruct student, Meek’s composite “monitorability”
570 (the mean of verbosity and faithfulness scores defined in [2]) rises after distillation almost entirely
571 via verbosity. The model looks more monitorable to that measure without being more monitorable to
572 the cue test. On the reasoning student, verbal acknowledgment falls without any matching reduction

Table 4: Per-text-cue acknowledgment, baseline R1-7B vs. W2SR (paired on question id). Ack drops on every text cue, and each cue’s discordant cells are one-directional (W2SR-only ack ≤ 1). The two non-text cues (`visual_squares`, `xml_metadata`) sit at 0% on both conditions and are omitted. Per-cue p -values are exploratory and uncorrected for multiplicity.

cue	baseline R1-7B	W2SR (1.5B)	paired Δ (n , disc. base/W2SR, p)
stanford_professor	15/32 = 46.9%	3/38 = 7.9%	-0.40 (30, 12/0, 5×10^{-4})
grader_hack	7/32 = 21.9%	1/38 = 2.6%	-0.20 (30, 7/1, 0.07)
insider_info	18/32 = 56.2%	2/38 = 5.3%	-0.50 (30, 15/0, 6×10^{-5})

in the cue’s behavioral pull — strengthened on the answer-only arm, not resolvable either way on the reasoning-preserving one. Compression alone predicts most of that raw collapse; what a compression-only account does not predict is the exploratory matched-length residual on the influenced = 1 subset — nor the Task K rerun, which removes most of the compression by fixing the supervision and still finds acknowledgment significantly below baseline. The safety-relevant component gets worse, not better.

The warning is sharp. An accuracy-only certification of any CoT-SFT student trained on MATH-style traces, including same-strength self-distillation rather than just weak-to-strong as in W2SR, can pass a model whose chain-of-thought has become markedly less revealing, and a monitorability check that rewards elaborate-looking CoT misses the same failure. The narrowness of the W2SR accuracy claim itself (in-family teacher and a base student, Section 4) sharpens this further. The certified guarantee is narrow on both axes.

7 Limitations

The headline is marginal at a matched budget. At the 8k budget used throughout most of this paper the acknowledgment drop is comfortable ($p = 0.005$); at the 32k budget that stops advantaging the trained arm it is $p = 0.043$ (§5.3). We report the latter. A reader should treat the effect as directionally consistent across both specifications and statistically marginal in the fairer one, and should not read the 8k figure as the estimate. The behavioural half is a wide null under both budgets, and the budget re-run establishes that widening the generation cap does not narrow it — the binding constraint is the number of questions, not the trace length.

The headline rests on a single arm and a single seed. The reasoning-preserving result (§5.2) is one training run of one teacher–student pair, scored by the headline judge alone, with no multi-judge battery on that arm. Its significance is also specification-sensitive: $p = 0.005$ as reported, $p = 0.077$ if we restrict to pairs where neither side hit the generation cap. The direction and magnitude are stable across those cuts, but the evidence is thinner than the surrounding robustness battery, which was measured on the answer-only arms. Re-running the strong-teacher and self-distillation arms with the corrected renderer is the obvious next experiment and we have not done it.

Most of our robustness evidence is attached to the answer-only arms. As described in Section 3.1, the student’s chat template removed the reasoning span from every SFT example in the original runs, so those arms were trained on the final-answer segments of math CoT traces rather than on the reasoning. That bounds what they show. Their CoT compression is explained by the supervision format and should not be read as an emergent property of CoT distillation. Their teacher-strength invariance is weaker evidence than it looks: with the teacher’s reasoning excluded from the loss, weak, strong, and self teachers differ only in their final answers, so their near-identical effect is close to what one would predict, and the invariance claim should be treated as untested under reasoning-preserving supervision. The multi-judge battery, the MMLU replication, the rubric swap and the CoT-conclusion analysis all sit on these arms too, so they corroborate the answer-only collapse directly and the headline only by analogy. What the answer-only arms do establish independently is that these models retain the cue’s behavioral pull — indeed strengthen it — while their traces stop reporting it. The certification argument survives on either arm, since answer-only SFT on math traces is itself a common and cheap post-training recipe, and it is exactly the kind of change an accuracy-only gate would wave through.

The behavioral comparison suffers severe differential missingness. This is the sharpest limitation on the paper and it bounds every influence number we report. Computing whether a cue moved the answer requires recovering an answer, and the arms differ enormously in how often they emit one: on the 150 matched (question, cue) pairs behind the answer-only comparison, 44.0% of baseline records have no recoverable answer against 6.7% of the trained arm’s. The cause is not subtle — the untrained baseline writes $\sim 20,000$ character traces into an 8k-token budget and frequently never concludes. Two consequences follow. First, the influence estimate is computed on a subset that discards nearly half of one arm and almost none of the other, and the discarded baseline records disproportionately contained text matching the cue target, so the direction of the extraction correction (§5) is partly what discarding them mechanically produces. Second, and more fundamentally, a comparison of “does the cue move the answer” between a model that answers 92% of the time and one that answers 56% of the time is not clean under any extraction rule: the loose rule buys coverage with records whose accuracy is at chance, and the strict rule buys accuracy with attrition. We report the strict rule, disclose the sensitivity, and note that the fix is not analytical but experimental — the evaluation needs a generation budget large enough that both arms actually finish. Until that is run, treat the acknowledgment results as the paper’s evidence and the behavioral results as directional.

Answer extraction is a measured quantity, not a given. Every behavioral number depends on recovering a final answer letter from a completion, and the arms differ enormously in how often they emit one within the generation budget (§5). We tightened the extractor to require an explicit answer marker after finding that a looser pattern was recovering cue restatements from unfinished baseline traces; that correction raised every reported effect. It also leaves the untrained baseline with no parseable answer on half its uncued items, which is why we decline to rest capability claims on uncued GPQA. A reader should treat the behavioral metrics as conditional on an extraction rule that we have documented and can be varied (LOOSE_FALLBACK_IS_AN_ANSWER in the released pipeline reproduces the pre-correction numbers).

The instruct arm only motivates the substrate choice. The faithfulness claim rests on the reasoning student. The matched-length residual depends on the cut we take, and we report all three. Pooled over all five cues the mid-length gap is closed (Task H bin 3, 17.5% vs 15.1%, Fisher $p = 0.80$) and the long-length gap is marginal (bin 4: OR ≈ 2.2 , two-sided $p = 0.14$); restricted to the three text cues, which removes a large across-arm imbalance in cue composition within bins, the gap is present at both lengths ($p = 0.068$ and $p = 0.016$) but on a W2SR side that does not share baseline’s question pool, and the question-matched versions keep the effect size while losing significance; on the influenced = 1 subset the mid-length gap is 70.0% vs 21.1%, OR ≈ 8.75 , $p = 0.003$, on $n = 10$ baseline records and conditioning on a post-treatment variable. Only the cue-stratified cut avoids both the composition confound and the post-treatment conditioning, and it is the strongest of the three, but it is supporting evidence only; none of them carry the argument on their own, which is why §5.2 exists. Compression accounts for a large share of the raw drop in absolute ack rates. We do not claim it is the sole mechanism, and we do not claim it is absent.

The degradation is shown within the Qwen R1-distill lineage, under weak, strong, and same-strength (self) teachers. A same-recipe cross-family test on R1-Distill-Llama-8B (self-distillation, matched LoRA config, 8k generation budget, same MATH L3 to L5 problem pool) fails the capability gate cleanly: GPQA accuracy goes from 50.0% to 22.5% (-27.5 pp at 0% empty completions on a verified-warm endpoint), at about $10.8\times$ CoT compression (15,420 to 1,424 chars median). This is substrate sensitivity, not infrastructure noise: the recipe that holds Qwen-7B capability while compressing about $14\times$ over-compresses Llama-8B and craters accuracy, putting the cross-family Self-B arm in the over-compression confound regime, so we do not run its monitorability eval. Cross-family generalization of the dissociation therefore remains open: at matched recipe, Llama fails the gate before the dissociation question is measurable.

The MMLU replication covers five STEM subjects; broader coverage, including humanities and social sciences, is untested, and the per-subject cells are small (30–40 cued samples each) even though the pooled comparison is well powered. Other limits: LoRA rather than full SFT; MATH-L3 to L5 SFT data only, with other domains untested; and judge robustness checked against two alternative families (gemini-2.5-pro, kimi-k2-0905) rather than a full panel. Absolute acknowledgment rates should be read as judge-dependent in any case [15]; what replicates across our three judges is the direction and the one-directional discordant pattern, not the level.

The influenced = 1 cut that carries the matched-length residual conditions on a post-treatment variable (whether the cue moved the answer), and it is one cell of a two-stratum by five-bin grid reported without multiplicity correction, with $n = 10$ on the baseline side. We report it because it is the subset monitorability is about, but it is the most fragile number in the paper and should be treated as a well-motivated exploratory finding rather than a confirmatory test. The Task K rerun (§5.2) now provides independent, intervention-based support for the same conclusion — a suppression component beyond compression — without conditioning on a post-treatment variable, though that rerun carries its own caveats (one arm, one seed, single judge).

Finally, the analysis plan was preregistered before any trained student existed (PREREGISTRATION.md). Several deviations should be noted. The preregistered primary estimand was W2SR — strong-teacher control, which is null here ($p = 0.09$); our headline is the preregistered secondary estimand, W2SR — baseline. Under the preregistered decision rules this is the H_0 case, “no weak-supervision-specific effect, but an SFT-on-CoT effect that is itself reportable,” which is what we report. A more consequential deviation: the preregistration’s validity gate required the W2SR student’s held-out MATH Pass@1 to exceed baseline by at least 5 absolute points (or recover at least 30% of the baseline-to-teacher gap) for condition-2 monitorability numbers to count, and the only committed gate measurement for the W2SR-weak arm — taken when it was re-gated alongside Task K — fails both branches (0.550 vs 0.695, with a repetition-loop flag; the weak teacher sits below the student’s baseline, so the gap-recovery branch does not apply), as does the Task K rerun’s own gate report (format-valid 0.80, gain +3.5pp). Under a strict reading of the prereg, the reasoning-arm monitorability numbers are therefore reported despite a failed capability gate. We report them because the paper’s claims do not rest on a capability gain — the finding is the collapse, measured on a substrate (GPQA) where the arm’s accuracy is preserved — but a reader holding us to the preregistered gate should treat condition 2 as failing its validity criterion and weight the reasoning-arm results accordingly. The prereg also called for judge–human agreement on a hand-labeled set before trusting the acknowledgment labels; we substituted cross-judge agreement across three model families and a within-judge rubric swap, which is weaker on the human-validity axis. Smaller deviations: the preregistered combined monitorability score is not reported for the reasoning-student arms (we report its components separately), the per-cue comparisons in Table 4 are exploratory and uncorrected, and the optional mixed-effects robustness model was not run.

8 Conclusion

Fine-tuning an R1-distill student on a weaker model’s math chain-of-thought, with the reasoning genuinely in the loss, cuts the trace’s acknowledgment of a planted cue from 23.6% to 16.6% at a matched 32k generation budget (paired $\Delta = -0.080$, $n = 150$, $p = 0.043$; the same comparison at 8k, where only the baseline is routinely truncated, reads 25.0% to 14.4% at $p = 0.005$) and at held-out MATH accuracy that does not fall, and with no detectable change in the cue’s pull on the answer (paired $\Delta = +0.048$, CI $[-0.060, +0.157]$, $p = 0.52$ — an underpowered null on a measurement degraded by differential answer emission, not a demonstrated invariance). Because the retrained traces are only ~27% shorter than the untrained model’s, the drop is not a by-product of the extreme compression that answer-only supervision produces; and because the manipulation is the training data rather than a filter on the outcome, the result does not condition on anything measured after treatment.

A second arm bounds how much of the effect trace length alone can produce. Due to a chat-template defect, an earlier run supervised the same student on the final-answer segments of the same traces; it compresses CoT $14\times$ and drives acknowledgment to 3.2%, so answer-only supervision reproduces a far larger collapse than reasoning-preserving supervision does. Length is nonetheless not the whole story even there: at matched length and matched cue composition the gap survives at OR between 2.7 and 4.7 across constructions, though underpowered wherever the question pool is held fixed (§5.1), and the MMLU replication reproduces the full collapse at $2.8\times$ compression rather than GPQA’s $14\times$. The collapse on that arm is also not specific to weak-to-strong distillation — a same-strength self-distillation control reproduces it at indistinguishable magnitude — and it survives a within-judge rubric swap (tightening to 9.4% $\rightarrow$ 0.0%) and two out-of-family judges. A same-recipe cross-family attempt on R1-Distill-Llama-8B fails the capability gate (-27.5 pp accuracy at $10.8\times$ compression), so the cross-family scope of the dissociation remains open. Separately, W2SR’s accuracy gain is real but conditional, requiring an in-family teacher and a base student.

725 Certifying any CoT-SFT student on accuracy alone, or on how elaborate its CoT looks, is insufficient
726 and potentially misleading. The cue-acknowledgment component of monitorability must be measured
727 directly.

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731 All errors are my own.

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